

Compiler Design Lab Programs

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Course Code: CSA1402

37. Write a LEX program to find the length of the longest word.

Code:

```
%{
#include <stdio.h>
#include <string.h>
int maxlen=0; char longest[256]="";
%}
%%

[a-zA-Z]+ {
    if(yyleng > maxlen){ maxlen = yyleng; strncpy(longest, yytext, sizeof(longest)-1);
longest[sizeof(longest)-1]=0; }
}

.\n    ;

%%

int main(){ yylex(); printf("Longest Word: %s\nLength: %d\n", longest, maxlen); return 0; }
int yywrap(){ return 1; }
```

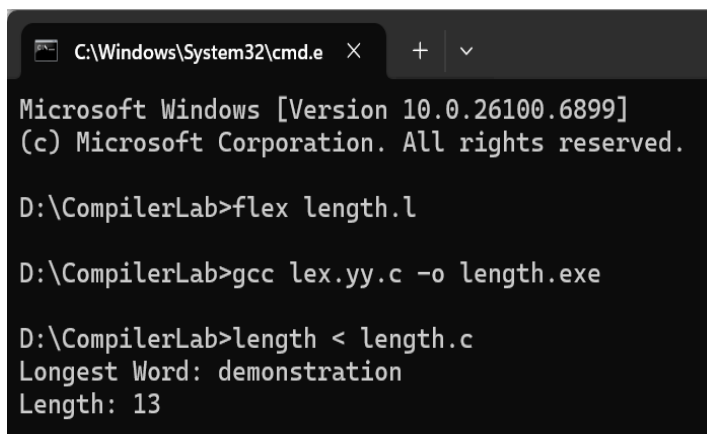
Output:

This is a demonstration example

Longest Word: demonstration

Length: 13

Output Screenshot:



```
C:\Windows\System32\cmd.e  X  +  v
Microsoft Windows [Version 10.0.26100.6899]
(c) Microsoft Corporation. All rights reserved.

D:\CompilerLab>flex length.l

D:\CompilerLab>gcc lex.yy.c -o length.exe

D:\CompilerLab>length < length.c
Longest Word: demonstration
Length: 13
```

38. Write a LEX program to count the frequency of the given word in a given sentence.

Code:

```
%{
#include <stdio.h>
#include <string.h>
char target[128];
int count=0;
}%
%%
[a-zA-Z]+ {
    if(strcmp(yytext, target)==0) count++;
}
.\n;
%%

int main(){
    printf("Enter the word to search: ");
    scanf("%127s", target);
    getchar(); /* consume newline */
    printf("Enter text (Ctrl+Z to end):\n");
    yylex();
    printf("Frequency of '%s': %d\n", target, count);
    return 0;
}

int yywrap(){ return 1; }
```

Output:

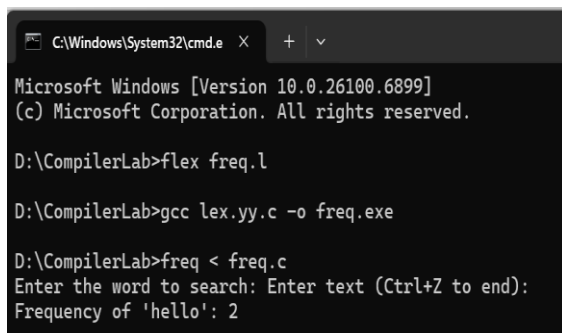
(When prompted) hello

Then text:

hello world hello everyone hello

Frequency of 'hello': 3

Output Screenshot:



```
C:\Windows\System32\cmd.e X + v
Microsoft Windows [Version 10.0.26100.6899]
(c) Microsoft Corporation. All rights reserved.

D:\CompilerLab>flex freq.l

D:\CompilerLab>gcc lex.yy.c -o freq.exe

D:\CompilerLab>freq < freq.c
Enter the word to search: Enter text (Ctrl+Z to end):
Frequency of 'hello': 2
```

39. Write a LEX code to replace a word with another word in a file.

Code:

```
%{
#include <stdio.h>
#include <string.h>
char oldw[128], neww[128];
%}
%%
[a-zA-Z]+ {
    if(strcmp(yytext, oldw)==0) printf("%s", neww);
    else printf("%s", yytext);
}
[ \t\n]+ { printf("%s", yytext); }
.      { printf("%s", yytext); }
%%

int main(){
    printf("Enter word to replace: ");
    scanf("%127s", oldw);
    printf("Enter new word: ");
    scanf("%127s", neww);
    getchar();
    printf("Enter text (Ctrl+Z to end):\n");
    yylex();
    return 0;
}
```

```
}
int yywrap(){ return 1; }
```

Output:

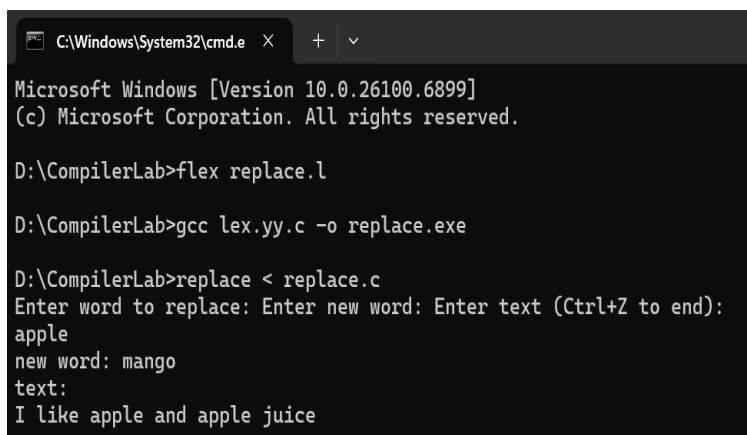
old word: apple

new word: mango

text:

I like apple and apple juice

Output Screenshot:



```
C:\Windows\System32\cmd.e
Microsoft Windows [Version 10.0.26100.6899]
(c) Microsoft Corporation. All rights reserved.

D:\CompilerLab>flex replace.l

D:\CompilerLab>gcc lex.yy.c -o replace.exe

D:\CompilerLab>replace < replace.c
Enter word to replace: Enter new word: Enter text (Ctrl+Z to end):
apple
new word: mango
text:
I like apple and apple juice
```

40. Write a LEX program to recognize a word and relational operator.

Code:

```
%{
#include <stdio.h>
%}

REL (<|=|>|=|==|!=|<|>)
WORD [a-zA-Z_][a-zA-Z0-9_]*

%%

{REL}    { printf("Relational Operator: %s\n", yytext); }
{WORD}    { printf("Word: %s\n", yytext); }
[0-9]+    { printf("Number: %s\n", yytext); }
.\n      ;

%%

int main(){ yylex(); return 0; }

int yywrap(){ return 1; }
```

Output:

a <= b

value > 100

Word: a

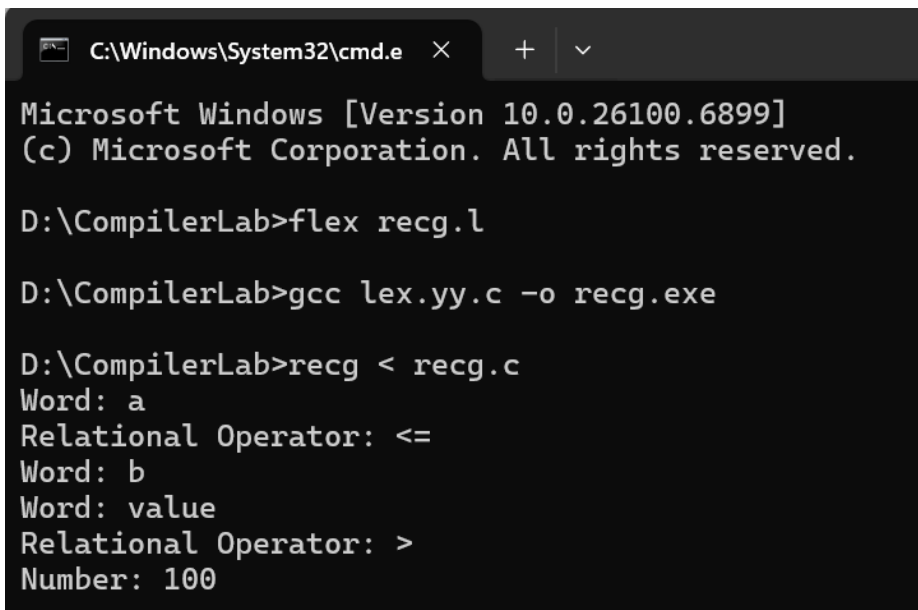
Relational Operator: <=

Word: b

Word: value

Relational Operator: >

Number: 100

Output Screenshot:

```
C:\Windows\System32\cmd.e × + v
Microsoft Windows [Version 10.0.26100.6899]
(c) Microsoft Corporation. All rights reserved.

D:\CompilerLab>flex recg.l

D:\CompilerLab>gcc lex.yy.c -o recg.exe

D:\CompilerLab>recg < recg.c
Word: a
Relational Operator: <=
Word: b
Word: value
Relational Operator: >
Number: 100
```